

BRMK557

Module 1

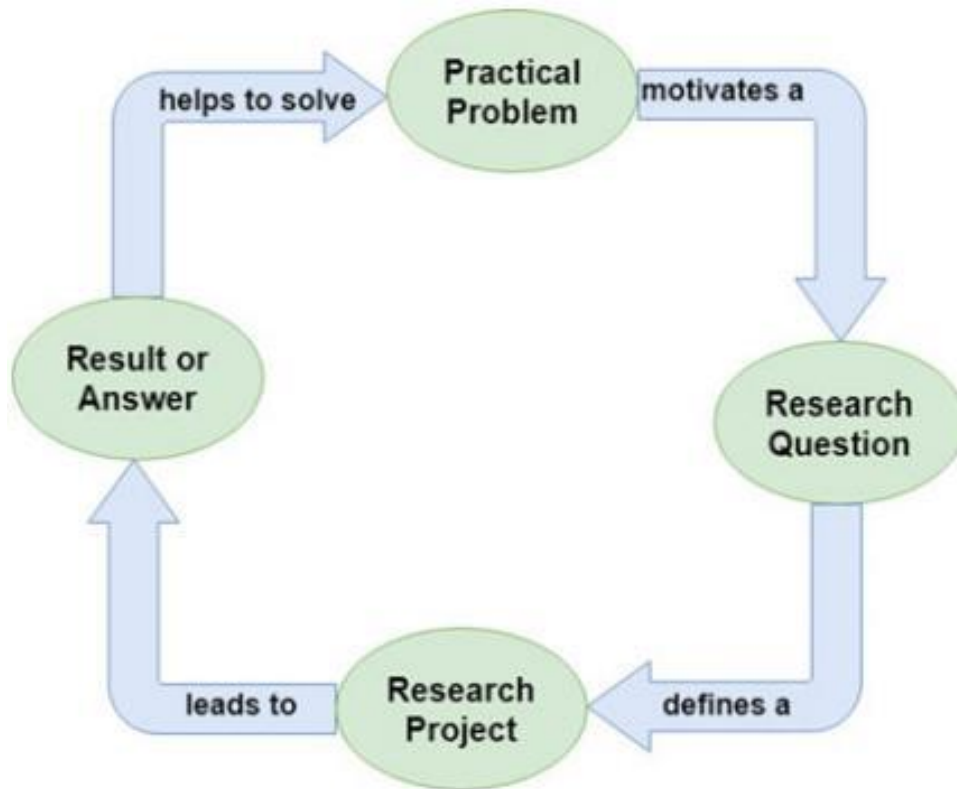
Module 1

- **Introduction:** Meaning of Research, Objectives of Engineering Research, and Motivation in Engineering Research, Types of Engineering Research, Finding and Solving a Worthwhile Problem.
- Ethics in Engineering Research, Ethics in Engineering Research Practice, Types of Research Misconduct, Ethical Issues Related to Authorship.

Research

- Research refers to a careful, well-defined (or redefined), objective, and systematic method of search for knowledge, or formulation of a theory that is driven by inquisitiveness for the unknown and useful on a particular aspect to make an original contribution to expand the existing knowledge base.
- Research involves formulation of hypothesis or proposition of solutions, data analysis, and deductions; and ascertaining whether the conclusions fit the hypothesis.
- Research is a process of creating, or formulating knowledge that does not yet exist.

Research Cycle



❖ Practical problem

one must be clear what the problem being attempted to solve
motivates a research question

❖ Research Question

helps to define a research project

❖ Research Project

An activity or set of activities that ultimately leads to result

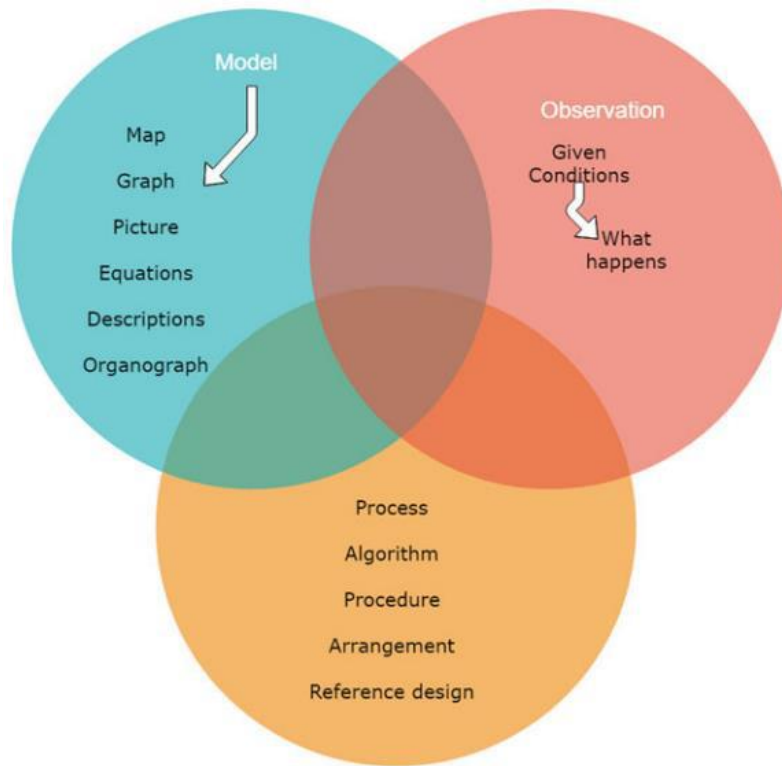
❖ Result

helps to solve the practical problem

The objective of a good research program is to try and gain insight into something.
Or indeed, to try and solve a problem.

Good research questions develop throughout the project actually and one can even keep modifying them.

Categories of accessing knowledge in research



The ways of developing and accessing knowledge come in three broad categories.

❖ **Observation**

❖ **Model**

❖ **Processes, algorithms, procedures, arrangements, or reference designs**

❖ **Observation is the most fundamental way of obtaining information from a source.**

❖ **It could be significant if the thing that we are trying to observe is strange or exciting, or is difficult to observe.**

❖ **Observation takes different forms from something like measurements in a laboratory to a survey among a group of subjects to the time it takes for a firmware routine to run.**

❖ **The observational data often needs to be processed in some form and this leads to the second category of knowledge, the model.**

Categories of accessing knowledge in research

- **Models**
- Models are approximated, often simplified ways of describing very complex interactions in the form of a statistical relationship, a figure, or a set of mathematical equations .
- For example, the modelling equation captures the relationship between different attributes or the behaviour of the device in an abstract form and enables us to understand the observed phenomena.
- The final category is a way of arranging or doing things through processes, algorithms, procedures, arrangements, or reference designs, to get a certain desired result.

Good research involves systematic collection and analysis of information followed by an attempt to infer a little bit of information beyond the already known information in a way that is a significant value addition.

Engineering Research

- Engineering research is a journey that traverses from a research area (example: Control Systems), to the topic (example: Control of Microbial Fuel Cells) and finally onto the problem (example: Adaptive Control of Single Chamber Microbial Fuel Cells) (Area → Topic → Problem).
- Getting a good problem to solve is more than half the work done. However, sometimes the journeys can be reverse, for example, the traversal from (Problem → Topic → Area).
- Engineering research is the process of developing the perspectives and seeking improvements in knowledge and skills to enable the recognition, planning, design, and execution of research in a wide range of forms relevant for engineering and technology investigations and developments.
- Can start off by describing some problem in the world that exists, that is bugging or worrying us and that it should be addressed
- The problem could be that there is something would like to do or accomplish but currently cannot because of lack of knowledge to do so.

Objectives of Engineering Research

- To solve new and important problems
- To develop new theoretical or applied knowledge and not necessarily limited to obtaining abilities to obtain the desired result.
- The objectives should be framed such that if incase of not able to achieve the desired result that is being sought, one can fall back to understanding why it is not possible, because that is also a contribution toward ongoing research in solving that problem.
- Of course, someone else might come along and propose a different approach where the desired objective is indeed possible to be achieved.

Motivation in Engineering Research

The possible motives may be the result of one or more of the following desires:

- Intrinsic motivations like interest, challenge, learning, meaning, purpose, are linked to strong creative performance
- Extrinsic motivating factors like rewards for good work include money, fame, awards, praise, and status are very strong motivators, but may block creativity. For example: Research outcome may enable obtaining a patent which is a good way to become rich and famous.
- Influences from others like competition, collaboration, commitment, and encouragement are also motivating factors in research. For example: my friends are all doing research and so should I, or, a person that I dislike is doing well and I want to do better.
- Personal motivation in solving unsolved problems, intellectual joy, service to community, and respectability are all driving factors.

Motivation in Engineering Research

The following factors would be a mix of extrinsic and intrinsic aspects:

- Wanting to do better than what has been achieved in the world
- Improve the state of the art in technology
- Contribute to the improvement of society
- Fulfilment of the historical legacy in the immediate sociocultural context.

Several other factors like government directives, funding opportunities in certain areas, and terms of employment, can motivate people to get involved in engineering research.

Government Agencies funding Research

- All India Council for Technical Education (AICTE)
- Council of Scientific and Industrial Research (CSIR)
- Defence Research and Development Organisation (DRDO)
- Department of Atomic Energy (DAE)
- Ministry of Ayush
- Department of Biotechnology (DBT)
- Ministry of Coal (MOC)
- Ministry of Earth Sciences (MoES)
- Department of Scientific and Industrial Research (DSIR)
- Indian Council of Medical Research (ICMR)
- India Meteorological Department (IMD)
- Indian Space Research Organisation (ISRO)
- Ministry of Electronics and Information Technology (MeitY)
- Ministry of Environment, Forest and Climate Change
- Ministry of New and Renewable Energy
- Ministry of Power
- Ministry of Social Justice & Empowerment (MOSJE)
- Ministry of Water Resources, River Development & Ganga Rejuvenation
- Ministry of Petroleum & Natural Gas
- University Grants Commission (UGC)

Types of Engineering Research

- **Descriptive versus Analytical**
- **Applied versus Fundamental**
- **Quantitative versus Qualitative**

Descriptive versus Analytical

- Descriptive research includes comparative and correlational methods, and fact-finding inquiries, to effectively describe the present state of art.
- The researcher holds no control over the variables; rather only reports as it is.
- Descriptive research also includes attempts to determine causes even though the variables cannot be controlled.
- In Analytical research, already available facts are analysis and critical evaluation are utilized.
- Some research studies can be both descriptive and analytical.

Descriptive versus Analytical

Examples of Descriptive Research

- Describe the current state of traffic congestion in a specific urban area.
- Market research surveys to understand consumer preferences
- Demographic studies to analyze population characteristics
- Observational studies of animal behavior in their natural habitat.

Examples of Analytical Research

- Analyze the factors contributing to traffic congestion and propose potential solutions.
- Analyzing medical records to identify risk factors for a disease
- Studying stock market data to predict future trends
- Investigating the causes of climate change by analyzing temperature and precipitation data.

Applied versus Fundamental

- Applied research seeks to solve an immediate problem facing the organization
- The primary objective of applied research is to determine a solution for compelling problems in actual practice.
- Research to identify social or economic trends, or those that find out whether certain communications will be read and understood are examples of applied research.
- Fundamental research is concerned with generalizations and formulation of a theory.
- Research concerning natural phenomena or relating to pure mathematics are examples of fundamental research
- Fundamental or Basic research aims in seeking information which could have a broad base of applications in the medium to long term

Applied versus Fundamental

Examples of Applied Research

- Developing a new drug or treatment protocol for managing diabetes in patients.
- Developing a new method to improve crop yields
- Finding ways to make products last longer
- Improving employee engagement in the workplace

Examples of Fundamental Research

- Investigate the molecular mechanisms underlying insulin resistance in diabetes to deepen our understanding of the disease
- Studying the fundamental nature of black holes in astrophysics
- Analyzing the structure of a protein
- Exploring the principles of quantum mechanics
- Examining the components of a cell

Quantitative versus Qualitative

- Quantitative research uses statistical observations of a sufficiently large number of representative cases to draw any conclusions.
- It aims to quantify relationships and patterns, making use of measurable variables.
- Surveys, experiments, and statistical analyses are common in quantitative research.

Examples

- A study measuring the effect of a new drug on patient recovery
- A survey assessing customer satisfaction with a product
- Qualitative researchers rely on a few non representative cases or verbal narrative in behavioural studies such as clustering effect in intersections in Transportation engineering to make a proposition
- Explore detailed feedback and experiences of engineering students using openended questions

Finding and Solving a Worthwhile Problem

- **Finding a Research Problem**
- A researcher may start with problems stated by the Supervisor or posed by others that are yet to be solved.
- Alternately, it may involve rethinking basic theories or formulating ideas from provided information or need to be formulated or put together from the information provided in a group of papers suggested by the Supervisor.
- Research scholars face the task of finding an appropriate problem to begin their research.
- **Skills Needed**
- Skills required for finding a research problem are crucial but often not explicitly taught. Critical thinking about possible implications is important.

Finding and Solving a Worthwhile Problem

Identifying a Problem:

- Once the problem is identified, the process of literature survey and technical reading takes place to further ascertain the significance and validity of the intended problem.
- An initial spark is ideally required before the process of literature survey to begin the research.
- The process may involve an initial spark from an oral presentation by somebody
- It is followed by asking questions or introspection about this perspective oral presentation.
- Developments in other subjects may produce a tool or a result which has direct implications to the researcher's subject and may lead to problem identification.

Finding and Solving a Worthwhile Problem

- **Attributes or characteristics of a Worthwhile Research Problem:**
- Once a potential research problem is identified, the researcher faces the critical task of evaluating its worthiness.
- A worthwhile research problem possesses one or more attributes, such as being non intuitive(something that goes against common intuition) or counter intuitive(does not align with what one might naturally expect based on prior knowledge or experience), even to someone familiar with the area.
- These attributes include:
 - Addresses a topic that the research community has been anticipating.
 - Simplifies a central part of theory.
 - Introduces a new result, initiating a new subject or area.
 - Offers a novel method or enhancements to existing methods with practical applications.
 - sometimes, provides a result that stops further work in a particular area.

Finding and Solving a Worthwhile Problem

- **Decision to Tackle a Problem:**
- The researcher has to be convinced that the problem is worthwhile before beginning to tackle it because best efforts come when the work is worth doing, and the problem and/or solution has a better chance of being accepted by the research community
- Not all problems that one solves will be great, and sometimes major advancements are made through solutions to small problems dealt with effectively.
- Some problems are universally considered hard and open, and have deep implications and connections to different concepts and most researchers in their lifetime do not get into such problems.
- The question a researcher has to grapple with whether the time investment is worth it given that the likely outcome is negative, and so it is a difficult personal decision to make.
- At the same time, even in the case of failure to solve the intended hard problem, there may be partial/side results that serve the immediate need of producing some results for the dissertation.

Finding and Solving a Worthwhile Problem

- George Pólya (1887–1985) suggested a 4-step procedure for mathematical problem-solving which is relevant to engineering researchers as well.
- Recent work also suggest the relevance of these recommendations.
- The recommended steps to solve a research problem are

(i) Understand:

- Understand the problem, and restate it in simple words as if it's your own
- visualize the problem by drawing figures, and determine if something more is needed.

(ii) Explore Strategies

- One must start somewhere and systematically explore possible strategies to solve the problem or systematically look for patterns that might lead to a solution.

(iii) Implement:

- Execute the plan to see if it works, and if it does not then start over with another approach.
- Having delved into the problem and returned to it multiple times, one might have a flash of insight or a new idea to solve the problem.

(iv) Reflect

- Looking back and reflecting helps in understanding and assimilating the strategy, and is a sort of investment into the future.

Ethics in Engineering Research

- Ethics generally refers to a set of rules distinguishing acceptable and unacceptable conduct, distinguishing right from wrong.
- Most people learn such norms in their formative years but moral development continues through different stages of growth.
- Although everyone recognizes common ethical norms, there can be differences in interpretation and application.
- International norms for the ethical conduct of research have been there since the adoption of the Nuremberg Code in 1947.
- The Nuremberg Code, is a set of ethical principles for human experimentation.
- It originated in response to the Nuremberg trials, where Nazi doctors were prosecuted for conducting inhumane experiments during World War II.
- Researchers must obtain informed consent from participants, especially in studies involving human subjects.

Ethics in Engineering Research

Influence of the British Royal Society on Research Credit:

- The issues associated with research credit find their roots in the seventeenth-century itself.
- British Royal Society (BRS), has been established with an aim to refine the methods and practices of modern science.
- BRS played a crucial role in shaping the timing and credit issues related to research results.
- According to this, priority for publication was given to whoever first submitted findings, rather than focusing on determining who had first made the discovery.

Ethics in Engineering Research

Ethical Considerations in Authorship:

- Two simple but significant questions to address the tricky issue of authorship in research are :
 - (i) who should be included as an author and
 - (ii) the appropriate order of listing authors.
- In today's interconnected world, the issue of co authorship is very relevant to all researchers, challenging the contributions during different phases of research.
- There are issues around individuals who may be actively involved in the research process but may not contribute to the drafting phase.
- Moreover, certain universities have imposed restrictions on co authorship and prevent malpractices

Research ethics and the responsible conduct of research

- Government bodies, and universities worldwide have adopted certain codes for research ethics.
- Research ethics and the responsible conduct of research are often erroneously used interchangeably.
- Research ethics examines the appropriate application of research outcomes, while responsible conduct of research deals with the way the work is undertaken.
- Research ethics focuses on ensuring integrity, honesty, and fairness in the research process.
- Key areas include the treatment of research subjects, confidentiality, data handling, and ethical communication of research outcomes.
- In essence, research ethics addresses the ethical implications of the research itself.

Research ethics and the responsible conduct of research

Responsible Conduct of Research(RCR) :

- Responsible Conduct of Research is a broader concept that extends beyond the ethical dimensions of the research itself.
- It encompasses the entire research process, emphasizing ethical behavior in interactions, collaborations, and the dissemination of results.
- RCR aims to maintain high standards of integrity and professionalism through out the research process.

Research ethics and the responsible conduct of research

- **Example: Consider a project focused on developing driverless cars.**
- During testing, researchers uncover vulnerabilities that could be exploited for malicious purposes.
- **Research ethics considerations in this scenario might involve:**
- (i) What ethical considerations should guide the development of such technology?
- (ii) How can the team prevent the misuse of autonomous vehicles for harmful activities?
- (iii) Are there ethical guidelines for the responsible development of autonomous systems
- **Responsible Conduct of Research challenges in this scenario involves**
- (i) Are researchers actively engaged in developing safety measures for autonomous cars?
- (ii) Do they collaborate with cyber security experts to secure the technology?
- (iii) How do they communicate potential risks to the public and regulatory bodies?

Research ethics and the responsible conduct of research

Research ethics addresses the broader implications and applications of the research outcomes, while the responsible conduct of research focuses on the integrity and methodology of the research process itself.

Ethics in Engineering Research Practice

- Ethics in engineering research practice extends beyond the laboratory or research setting to encompass the ethical challenges faced by engineers in the practical application of their knowledge and skills.

Ethical Concerns in Technological Developments:

- Technological developments raise a whole range of ethical concerns such as privacy issues and data related to surveillance systems.
- Engineering researchers need to make ethical decisions and are answerable for the repercussions borne out of their research as outcomes.
- Engineering researchers bear the responsibility of making ethical decisions, and they are answerable for the consequences arising from their research outcomes.

Types of Research Misconduct

- Engineering research should be conducted to improve the state-of-the-art of technologies.
- Research integrity encompasses
 - dealing fairly with others
 - honesty about the methods and results
 - replicating the results wherever possible to avoid errors
 - protecting the welfare of research subjects
 - ensuring laboratory safety, and so forth
- To prevent mistakes, peer reviews should take place before the research output is published.
- Summary of different types of research misconduct as described in research articles
- Fabrication (Illegitimate creation of data),
- Falsification (Inappropriate alteration of data)
- Plagiarism (Taking others work)

Types of Research Misconduct

- **Fabrication (Illegitimate creation of data):**
- This involves the act of creating data or experiments with preconceived notions about the expected outcomes.
- This unethical practice may arise when there are time constraints imposed by supervisors or customers leading researchers to generate data rather than waiting for genuine results
- **Example: Imagine a student conducting a science experiment on the growth of plants under different light conditions.**
- Due to a lack of time, the student decides to fabricate the data by recording measurements that were never actually taken.
- The fabricated data might show consistent and impressive growth differences between plants subjected to various light conditions.
- In this case, the fabrication involves making up experimental results instead of honestly recording the actual outcomes of the plant growth experiment.
- This kind of behavior is unethical and goes against the principles of honesty and integrity in research.

Types of Research Misconduct

- **Falsification (Inappropriate alteration of data)**
- Falsification involves the inappropriate alteration of data or experiments, including misrepresentation, misinterpretation, or illegitimate changes to support a desired hypothesis.
- This unethical practice occurs even when the actual data obtained from experiments indicate a different outcome.
- Falsification undermines the credibility and reliability of scientific research by presenting distorted information to align with a preconceived notion or agenda
- Plagiarism (Taking others work)

Types of Research Misconduct

- **Plagiarism (Taking others work)**
- Plagiarism is defined as the act of using or reusing someone else's work, including text, data, tables, figures, illustrations, or concepts, without proper attribution.
- It involves presenting the work as if it were one's own without explicit acknowledgment.
- The concept of self-plagiarism occurs when researchers verbatim copy or reuse their own previously published work without appropriate citation.
- This practice is considered unacceptable in scientific literature